

Theory Section

1: Missing Value Imputation: -

Missing value imputation is the process of replacing missing data with estimated values. It helps improve data quality and ensures that machine learning algorithms can work effectively without losing important information.

2: Simple Imputer: -

Simple Imputer is a technique used to replace missing values using statistical measures such as mean, median, or most frequent value. It is easy to implement and commonly used for basic preprocessing.

3: Missing Indicator and Random Sample Imputation: -

A Missing Indicator creates a new feature that identifies whether a value was originally missing. Random Sample Imputation replaces missing values with randomly selected existing values from the same feature, preserving the original data distribution.

4: KNN Imputer: -

K-Nearest Neighbors (KNN) Imputer fills missing values by using information from similar records. It identifies the nearest neighbors based on feature similarity and estimates missing values accordingly. This method often provides better accuracy than simple imputation techniques.

5: MICE (Multiple Imputation by Chained Equations)

MICE is an advanced imputation technique that predicts missing values using relationships among multiple variables. It performs multiple iterations to generate more realistic and accurate estimates, making it suitable for complex datasets.

5: Outlier Detection

Outliers are observations that differ significantly from the rest of the data. They may occur due to measurement errors, data entry mistakes, or genuine rare events. Detecting and treating outliers is important because they can negatively affect statistical analysis and machine learning models.

7: Z-Score Method

The Z-Score method measures how many standard deviations a data point is away from the mean. Observations with an absolute Z-Score greater than 3 are generally considered outliers.

8: Interquartile Range (IQR) Method

The IQR method detects outliers using the spread of the middle 50% of the data. Values below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$ are considered outliers.

9: Percentile Method

The Percentile Method identifies outliers by examining extreme values at the lower and upper ends of the distribution. Commonly, the 1st and 99th percentiles are used as thresholds.

10: Winsorization

Winsorization is an outlier treatment technique that replaces extreme values with specified percentile limits instead of removing them. This preserves the size of the dataset while reducing the impact of extreme observations.

Summary

In this project, multiple missing value imputation techniques and outlier detection methods were applied to improve data quality. The cleaned dataset is more reliable, consistent, and suitable for machine learning applications such as disease risk prediction.